SDG BLOCKCHAIN ACCELERATOR

Impact Measurement Framework

1. Project Information

Project Name
Boosting El Salvador's Circular Economy through Blockchain-Based Plastic Recovery Traceability

Challenge & UNDP Office
Solid waste management remains largely informal in El Salvador, with low recycling rates, fragmented value chains, and no standardized mechanism for traceability or Extended Producer Responsibility (EPR) enforcement. This results in limited accountability, under-monetized recycling activity, and weak data for policy and climate reporting.

UNDP Office: El Salvador

2. SDG Alignment

This pilot is structured to deliver measurable contributions across environmental sustainability, economic inclusion, and digital innovation.

SDG 12: Responsible Consumption and Production
The project directly contributes to reducing waste leakage by introducing traceable recovery flows and verifiable recycling outputs.

- **Target 12.5:** Reduce waste generation through recycling and reuse
- **Indicators:**
    - Total tons of plastic waste recovered and verified
    - Number and volume of Plastic Credits issued
    - Percentage of recovered plastic successfully processed vs collected

SDG 8: Decent Work and Economic Growth
The initiative formalizes informal waste workers by integrating them into a digital system where recovery is tracked and monetized.

- **Target 8.3:** Promote formalization and productive employment
- **Indicators:**
    - Number of recyclers onboarded and active
    - Average income increase per recycler (baseline vs pilot)
    - Frequency of recycler participation (transactions per user)

SDG 9: Industry, Innovation and Infrastructure
The deployment of blockchain infrastructure introduces transparency and interoperability into the waste management ecosystem.

- **Target 9.4:** Upgrade infrastructure and adopt clean technologies

- **Indicators:**
  - Deployment and uptime of blockchain-based traceability system
  - Number of integrated actors (collectors, aggregators, verifiers, buyers)
  - System usage metrics (transactions, batches recorded)

SDG 13: Climate Action
Plastic recovery contributes indirectly to emissions avoidance by reducing landfill use and promoting circular material flows.

- **Target 13.2:** Integrate climate measures into strategies
- **Indicators:**
  - Estimated CO_2 equivalent avoided through verified recovery
  - Volume of plastic diverted from landfill or open dumping
  - Availability of auditable environmental data for reporting

3. Key Performance Indicators (KPIs)

The pilot defines both output and outcome-level KPIs to ensure a comprehensive view of impact:

Operational KPIs

- Plastic waste tokenized: **300–400 tons** during pilot
- Plastic Credits issued: **Initial issuance completed within pilot phase**
- Number of traceable batches recorded: **All recovery flows digitized (target: 100%)**
- System uptime: **≥95% availability of platform and services**

Ecosystem KPIs

- Recyclers onboarded: **150–200 individuals**
- Active recyclers (monthly): **≥70% of onboarded users**
- Accredited waste actors: **2–3 aggregators/processors**
- Stakeholder institutions engaged: **≥5 (UNDP, local authorities, private sector, NGOs)**

Adoption & Capacity KPIs

- Training sessions delivered: **Multiple modules across onboarding phases**
- Training completion rate: **≥80% of participants**
- User retention rate: **≥60% over pilot duration**

Impact KPIs

- Traceability rate: **100% of tokenized plastic tracked end-to-end**
- Income uplift for recyclers: **Measured vs baseline (qualitative + quantitative)**
- Volume of plastic diverted from landfill: **Aligned with tokenized volume**

4. Tracking Methods

The measurement framework relies on a hybrid architecture combining blockchain-based verification with operational data systems.

On-chain Metrics (Source of Truth)
These metrics are recorded directly on blockchain and provide immutable, auditable data:

- Plastic Credits minted, transferred, and retired
- Unique batch IDs representing recovered plastic
- Wallet-level activity (recyclers, aggregators, buyers)
- Timestamped records for each step in the lifecycle (collection → aggregation → verification → issuance)

Off-chain Metrics (Operational Context)
These metrics provide depth and context not captured on-chain:

- Recycler onboarding profiles (demographics, location, activity level)
- Training attendance and certification records
- Institutional engagement logs (meetings, agreements, participation)
- Payment and disbursement records (if not fully on-chain)

Data Validation Approach

- Cross-referencing on-chain batch records with physical verification checkpoints
- Use of verifiers or auditors to confirm recovery claims before token issuance
- Periodic reconciliation between system data and field reports

5. Baseline vs Projected Outcomes

Baseline (Pre-Pilot)

- No standardized digital traceability for plastic recovery
- No Plastic Credit issuance mechanism
- Informal recycler ecosystem with limited income visibility
- Minimal data available for policy, reporting, or climate accounting

Projected Outcomes (During Pilot)

- **300–400 tons** of plastic fully traceable through the system
- First operational Plastic Credit issuance cycle completed
- Real-time dashboard providing visibility into recovery flows and impact
- Digitally onboarded recycler network with improved income opportunities
- Increased trust and transparency across stakeholders

Post-Pilot Trajectory (Expected)

- Expansion of traceability system to additional regions or waste streams
- Integration with EPR schemes and corporate buyers of Plastic Credits
- Standardization of data for national and international reporting

6. Monitoring & Reporting Plan

A structured monitoring system ensures continuous visibility and accountability across stakeholders.

Frequency

- Quarterly formal reporting
- Continuous real-time monitoring via dashboard

Responsible Parties

- **Plastiks:** Platform operations, data collection, reporting consolidation
- **UNDP El Salvador:** Oversight, validation, stakeholder coordination

Reporting Outputs

- **Live Dashboard:**
  - Real-time KPIs (plastic recovered, credits issued, active users)
- **Quarterly Reports (PDF):**
  - KPI performance vs targets
  - Narrative analysis and key insights
  - Challenges, risks, and mitigation updates
- **Blockchain Logs:**
  - Verifiable transaction records accessible for audit

Data Governance

- Defined ownership of on-chain vs off-chain data
- Access controls for sensitive user or institutional data
- Transparency layer for public-facing metrics

7. Risks & Mitigation

Low Technology Adoption
Recyclers may face barriers in using digital tools.

- *Mitigation:* Simplified UX, mobile-first interfaces, in-person onboarding, continuous support

Institutional Coordination Gaps
Misalignment between stakeholders may slow implementation.

- *Mitigation:* Early engagement, clearly defined roles, formal MoUs, regular coordination checkpoints

Financial Sustainability
Dependence on demand for Plastic Credits could impact long-term viability.

- *Mitigation:* Early identification of buyers, integration with ESG and corporate sustainability pipelines

Data Integrity Risks
Potential discrepancies between reported and actual recovery.

- *Mitigation:* Blockchain immutability, verifier checkpoints, batch-level validation processes

Operational Constraints
Logistics, infrastructure, or local conditions may affect execution.

- *Mitigation:* Flexible pilot design, phased rollout, local partnerships